

WASP-2b

R-1401

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_210812 · created 2026-08-01T04:09:28+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459438.7938 +/- 0.0066 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.075 +/- 0.046
Transit depth (Rp/Rs)^2	0.57 +/- 0.7 [%]
Orbital Inclination (inc)	82.66 +/- 2.88
Airmass coefficient 1 (a1)	1.35 +/- 0.23
Airmass coefficient 2 (a2)	-0.29 +/- 0.59
Scatter in the residuals of the lightcurve fit is	16.62 %
Best Comparison Star	#2 - [74, 205]
Optimal Aperture	5.2
Optimal Annulus	6.48
Transit Duration (day)	0.031 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.075 ± 0.046 vs 0.1326 (deviation 0.73σ)
Duration fit vs published	0.031 d vs 0.0737 d (deviation 0.73σ)
Mid-time O–C	6.6 min
Depth SNR	1.0
Residual scatter	16.62 %

Light curve

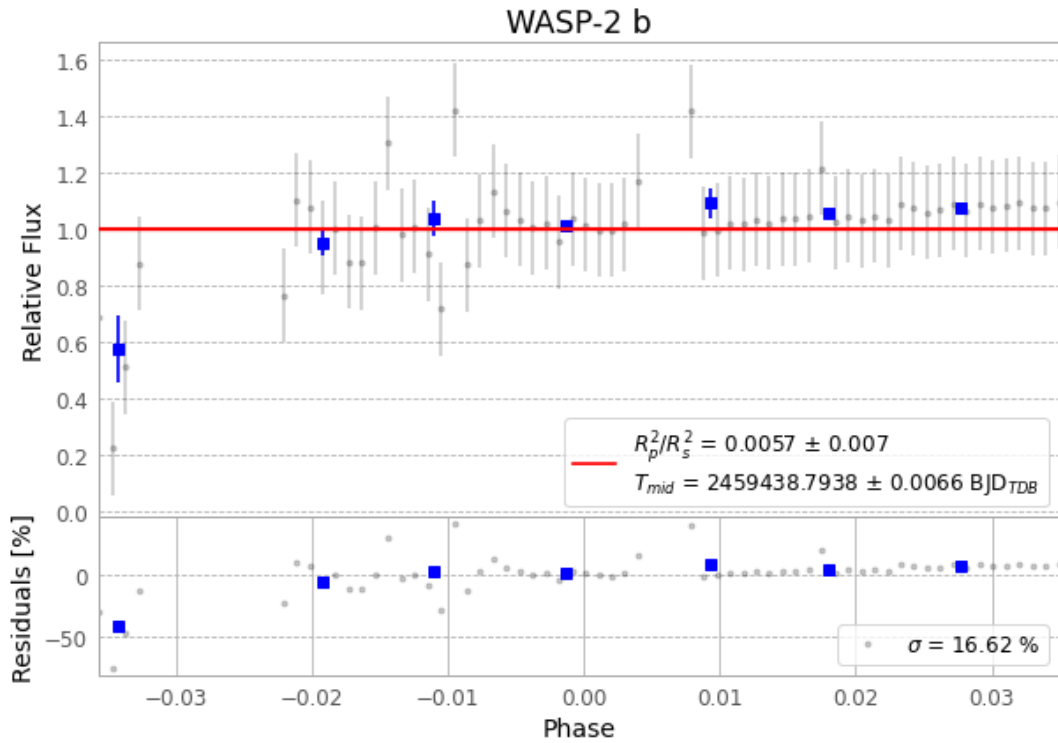


Figure 1 — FinalLightCurve_WASP-2 b_2021-08-11.png (SHA-256 ee476a7f7e7efc43...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [384, 119], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 60bb5b8133d2b0c6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

74 FITS frames (49 MB), WASP-2210812050312.FITS → WASP-2210812084212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-210812.log (d38a1289206e...), FinalLightCurve_WASP-2 b_2021-08-11.png (ee476a7f7e7e...), FinalLightCurve_WASP-2 b_2021-08-11.csv (9dc4780e7fa8...)

Compute resources

Wall time 215.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)